

## **PHYS580 Lab 2**

### **Assignment:**

Go to the course brightspace page and download the starter programs for Lab 2. Make copies before editing them to work on the assignment. (Or use them as guidance to write corresponding programs in your chosen languages.) You will work on the projectile motion today. One of the provided programs computes the cannon shell trajectory using the isothermal model as discussed in class.

1. First, for a cannon shell projectile, take a fixed angle of projection, say,  $45^\circ$ , and the initial velocity, say, 700 m/s, and graphically show the differences in trajectories for the following different models: (a) gravity only, no air drag, (b) air drag with constant air density, (c) air drag with the isothermal model of air density, and (d) air drag with the adiabatic model, using  $\gamma = 1.4$ . Set  $B_2/m = 4 \times 10^{-5}$  (in 1/m units) at ground level. The adiabatic model is where  $|F_{\text{drag}}| = B_2 (1 - a y/T_{\text{grd}})^{1/(\gamma-1)} v^2$  ( $a$  is the rate of temperature decrease per rise from sea level. Use  $a = 6.5 \times 10^{-3}$  K/m and  $T_{\text{grd}} = 293$  K). [Can you derive/justify this form of  $F_{\text{drag}}$ ?] Give some arguments for why the shapes and ranges of these 4 trajectories may be expected, comparatively.
2. Second, for each of the cannon shell projectile models you implemented in (1), program any reasonable procedure that i) computes the angle of projection that gives maximal range and ii) obtains that maximum range. [Use of the bisection method is advised.] Take the parameters as in (1), and compute the optimal angle to within 1/10 of a degree and estimate the corresponding accuracy of the computed maximum range. Compared with these, is your choice of  $\Delta t$  justified?